

Strategic Workforce & Employee Attrition Analytics

Data Analytics Portfolio Executive Presentation Document

1. Executive Summary

This presentation-grade report serves as the strategic synthesis of the data cleaning pipeline (`.ipynb`) and interactive dashboard interface (`.twbx`) compiled for the Employee Attrition Analysis Project. The objective is to convert complex HR matrices into clean, structured insights for senior leadership decision-making. Through structured pipeline filtering, anomalous variance metrics were successfully removed, allowing the isolation of exact organizational flight risk indicators.

1,470	16.12%	30.53%
TOTAL CORPORATE ROSTER	BASELINE ATTRITION RATE	OVERTIME TURNOVER RATE

2. Data Engineering & Pipeline Infrastructure

The initial backend architecture handled comprehensive auditing and normalization inside the Jupyter Notebook pipeline before staging data for the frontend Tableau Public environment. Key procedures implemented include:

- **Zero-Variance Column Removal:** Dropped uninformative variables (e.g., constants or identifiers) that lacked empirical variation.
- **Logical Contradiction Filtering:** Screened records for operational consistency across roles, departments, and tenure metrics.
- **Data Dictionary Synthesis:** Formatted fields into explicit Dimensions and Measures to guarantee dynamic interactive tracking.

3. Key Analytical Insights Unlocked

Core HR Dimension	Statistical Metric / Behavior Identified	Strategic Impact Level
Overtime Status	Employees working overtime log 30.53% attrition vs. 10.44% baseline.	Critical Risk Factor
Marital Demographics	Single cohort exhibits the highest group-level exit velocity at 25.53%.	High Mobility Risk
Compensation Disparity	Departed staff averaged \$4,787 monthly income vs. \$6,832 for retained staff.	Structural Financial Gap
Job Level 2 Paradox	Mid-tier personnel experience low overtime payouts relative to heavy labor output.	High Burnout Risk

Strategic Dashboard Visual Fix Note

The X-axis layout container tracking **Income Disparity across Job Levels and Overtime Status** was optimized by increasing the structural vertical padding. This adjustment fully resolved text truncation, ensuring complete alignment and visibility of nested metrics across all four job tiers.

4. Leadership Action Plan & Recommendations

1. **Implement Overtime Caps & Audits:** Formulate an operational limit on mandatory overtime hours, initiating a management review whenever a specific team consistently logs over 15% surplus hours.
2. **Correct Job Level 2 Compensation Tiers:** Introduce structured performance-based bonuses or adjust baseline banding for Level 2 roles to eliminate competitive poaching and close the retention salary gap.
3. **Deploy Targeted Engagement Metrics:** Leverage Tableau Public action filters to continuously monitor vulnerable demographic subsets, introducing proactive retention check-ins before critical turnover intervals.